IIT ROORKEE

FREE ONLINE EDUCATION swayam शिक्षित भारत, उन्नत भारत

NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 37: Material Flow Analysis: Case Study

By-
Prof. Bhola Ram Gurjar
Dept. of Civil Engineering



Contents

Case study: MFA based projections of material flows and stocks in the urban transport sector until 2050 – A scenario-based analysis for the city of Vienna, Austria

- Study area
- Background & need of study
- Research objectives & questions
- Scenarios for analysis
- Material considered for the study
- Projections for scenarios
- Traffic area & rail infrastructure projections
- Material input/output flow
- Conclusions

Study area: Vienna, Austria

- Austria is situated in Europe
- Vienna is the capital city of Austria
- It is the most populous city in Austria with population about 2 million.
- Vienna has robust public transport system well connected with rest of the country.



Background of the study

- In cities, transport sector significantly contributes to global resource and energy consumption
- Decarbonizing transport sector is increasingly recognized as an important means of mitigating climate change
- Studies have focused almost exclusively on energy and CO_2, thereby neglecting the consumption of material resources required to transform urban transport systems to be compatible with a low carbon future

10-20% of Europe's urban population is exposed to excessive levels of nitrogen dioxide (NO_2), along with several other air pollutants. NO_2 pollution is highest along busy roads.

Need of the study

- Target to reduce Vienna's per capita CO_2-eq emissions in the transport sector by 50% by 2030 and by 100% by 2050.
- Propulsion technologies (EV) and a shift towards more environment friendly transport modes (walking, public transport)
- A shift in transport mode choice versus technological development often represent counter-poles during decision making. An optimum scenario required which benefits the most.

Objectives of the study

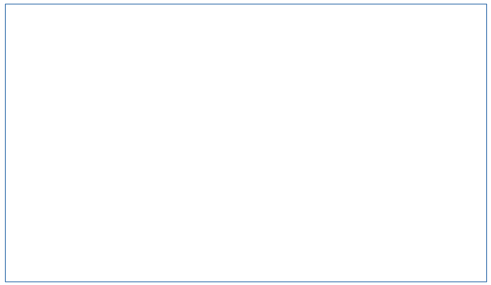
- Effects of transforming urban transport systems on future material stock and material flows using the city of Vienna as a case study.
- Material flow analysis for the infrastructure and vehicles required until 2050 taking different scenarios into consideration
- Scenarios are mainly characterized by different modal splits
- The time span of investigation covers the period from 2016 to 2050



Image: Transport modes in Vienna

Research questions

- How will the transport infrastructure and the vehicle fleet develop up to the year 2050 based on different future scenarios?
- How do different developments in the transport system affect material stocks and material flows?
- Are there saving potentials for raw materials within the future transport system alternatives and, if so, how can they be realized?



System Boundary

- The system under investigation is the transport system within the administrative border of Vienna.
- The transport infrastructure and vehicles are considered.
- A distinction is made between different modes of transport, namely motorized individual transport (MIT), non-motorized individual transport (NMIT) and public transport (PT).



Image: Administrative boundary of Vienna

Material Considered for analysis

- The materials are expressed in metric tons (t) and their multiples (e. g. Million tons (Mt)).
- The material groups considered are asphalt & bitumen, aluminum, batteries, brickwork, concrete, copper, glass, gravel & sand, iron & steel, rubber, plastics and wood.

Material Flows & Stocks considered 1/2

- Material flows related to the construction and demolition of transport infrastructure.
- Flows caused by the maintenance of transport infrastructure are also considered.
- Annual input and output flows are calculated according to the respective changes in stocks (increase/decrease).

Material Flows & Stocks considered 2/2

- The material flows necessary for maintenance are based on the renewal rates and useful lives of the infrastructures and infrastructural components
- Material flows associated with vehicle maintenance and operation are not part of the investigation



Image: Railway network infrastructure in Vienna

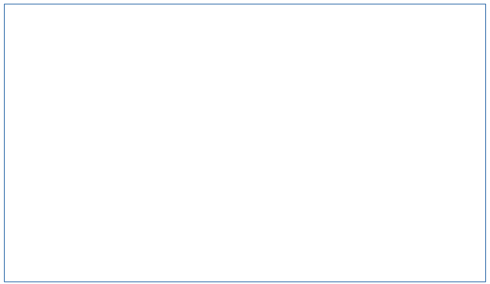
Scenario: A (Business as usual)

- 2016-30:
  - Modal split (MIT 25%, PT 38%, NMT 37%)
  - Infrastructure development - already planned network extensions and vehicle procurements

- 2030-50:
  - Projection of the prevailing system into the future
  - EV technology of the vehicle fleet remains at the current state (as baseline 2016)

Scenario: B (EV fleet)

- 2016-30:
  - Same as A except inclusion of EV in car fleet
- 2030-50:
  - Car share (25%)
  - Changed vehicle fleet
  - Replacement of fossil fueled cars by alternatives (electric, hydrogen)
  - Increase in the share of PT (45% from 38%)
  - Decrease in active mobility - NMT (30% from 37%)



Scenario: C (Public Transport)

- 2016-30:
  - Same as A except inclusion of EV in car fleet

- 2030-50:
  - Increase in the share of PT (55% from 38%)
  - Decrease in the share of MIT (<10%)
  - Changed vehicle fleet by EV/hydrogen
  - Constant development of the share of active mobility - NMT (35%)

  MIT: Motorized individual transport

Scenario: D (Active Mobility)

- 2016-30:
  - Same as A except inclusion of EV in car fleet
- 2030-50:
  - Increase in distances covered through active mobility - NMT (>45% from 37%)
  - Increase in the share of PT (45% from 38%)
  - Decrease in the share of MIT (<10%)
  - Changed vehicle fleet by EV/Hydrogen

Traffic mode projections

Transport service provided	Unit	2020	2030	2050 A_{BAU}	2050 $B_{+ BEV}$		2050 $C_{+ PT}$	2050 $D_{+ AM}$
Cars	(Million vehicle km/year)	4,670	4,960	5,470	6,240		1,680	1,680
Small lorry vehicles (<3,5 t)	(Million vehicle km/year)	510	580			720		
Heavy lorry vehicles (3,6-40t)	(Million vehicle km/year)	360	390			440		
		Public transport service						
Metro	(Million train km/year)	17.2	19.1	20.7	21.3		21.3	21.3
	(Number of trains in use/day)	160	150	190	170		230	230
Tram	(Million train km/year)	24.7	25.9	27.7	27.9		28.8	28.8
	(Number of trains in use/day)	470	520	530	480		750	750
Regional train	(Million train km/year)	8.4	8.4	8.9	8.2		12.9	12.9
	(Number of trains in use/day)	90	90	90	90		130	130
Public bus	(Million bus km/year)	39.5	41.9	44.9	44.9		49.4	46.9

Table: Transport services provided within Vienna in different scenarios

- Highest trips through personal vehicle is in scenario B and least in C & D and PT in scenario C.
- From year 2020 to 2030, no major improvement is achieved in terms of share of NMT/PT.

Share of vehicles (based on type)

Propulsion technology	2020				2030				2050			
	Car	Motorcycle	Lorry <3.5t	Lorry >3.5t	Car	Motorcycle	Lorry <3.5t	Lorry >3.5t	Car	Motorcycle	Lorry <3.5t	Lorry >3.5t
Petrol	45%	97%	5%		30%	85%			3%	5%		
Diesel	52%	1%	93%	100%	43%		77%	85%	3%			
Battery electric vehicle	<1%	2%	1%		15%	15%	17%	8%	82%	95%	95%	50%
Compressed Natural Gas (CNG)	<1%		1%		1%		2%	2%	1%			
Hybrid Petrol/Electro (hybrid)	<2%				5%		2%	2%	1%			
Diesel/Electric (hybrid)	<1%				5%		2%	2%	1%			
Hydrogen					1%		1%	1%	10%		5%	50%

- A drastic jump is expected in no. of EVs from 2020 to 2030 & 2050.
- EVs will also be used in commercial transport of goods.
- 95% motorbikes will be battery based by 2050.

Projection for Active Mobility

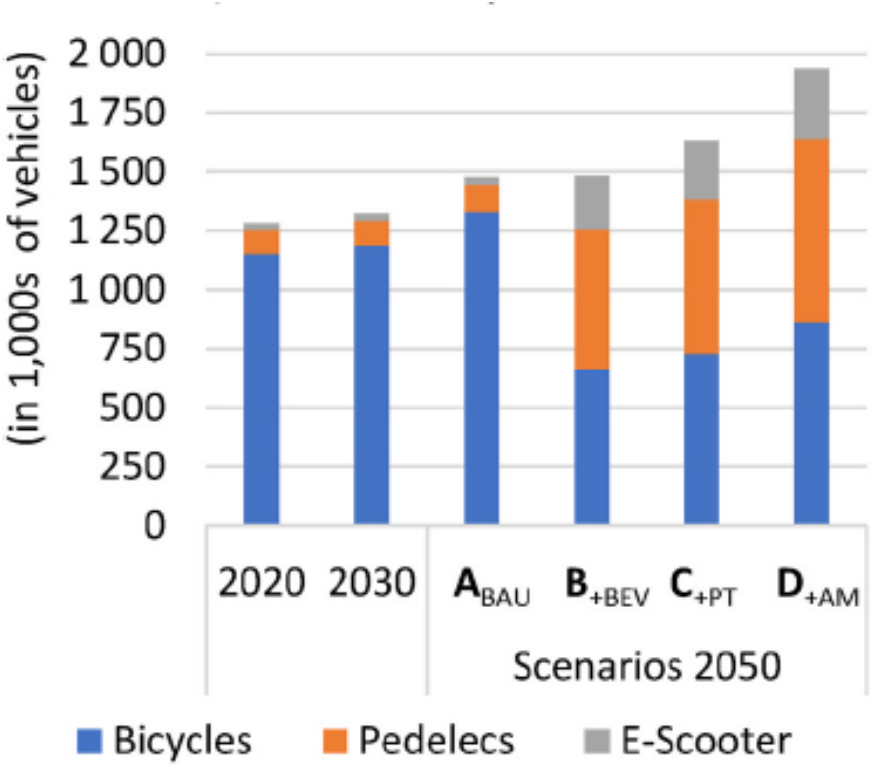


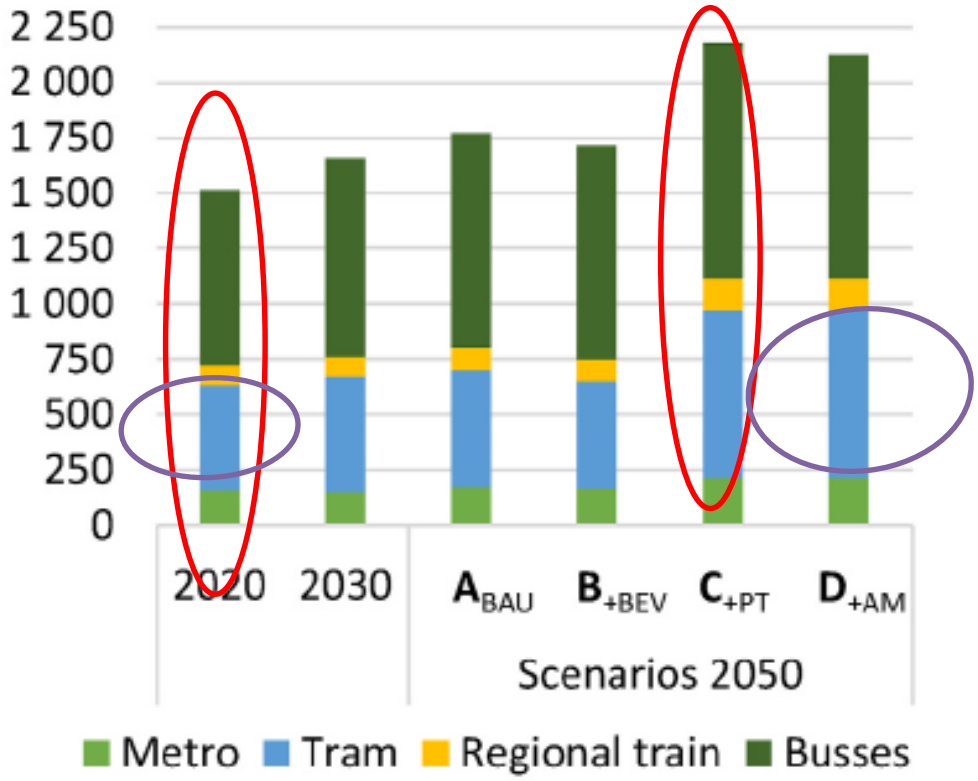


Image: Pedelecs Is E-bicycle

Image source: medfarmres.com

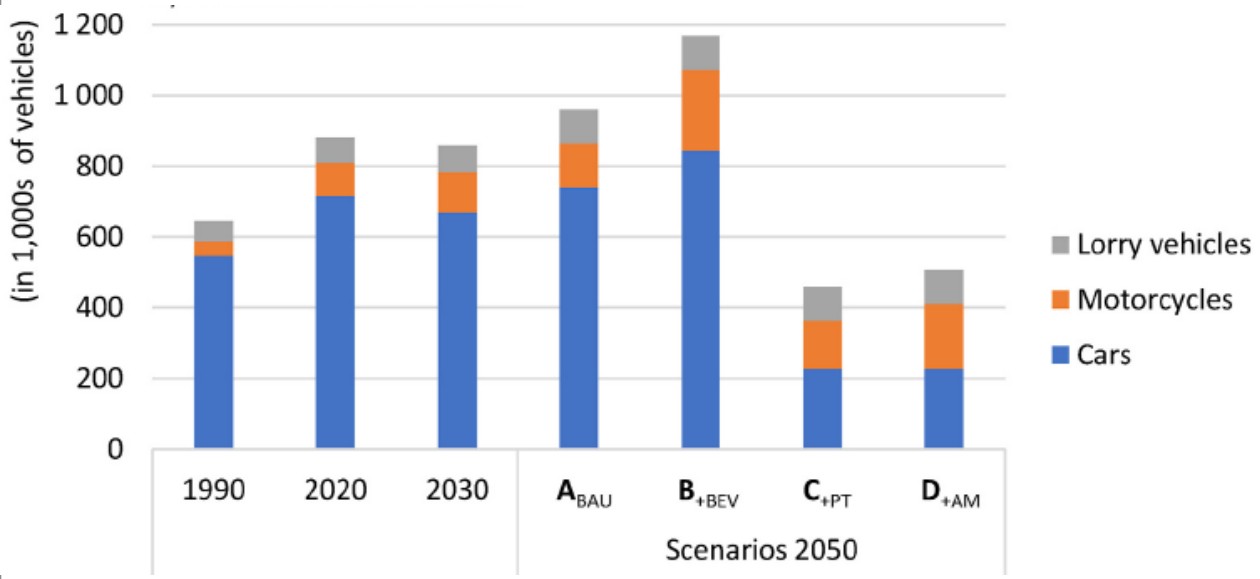
- Share of bicycles reduced in scenario B, C & D compared to A and 2030; replaced by pedelecs.
- Steady growth is projected in 2050 scenarios except A.

Projection for Public Transport



- In best case scenario for PT, it will grow drastically in 2050 compared to 2020.
- Share of tram in PT is expected to increase up to 47% in scenario D from 33% in 2020.

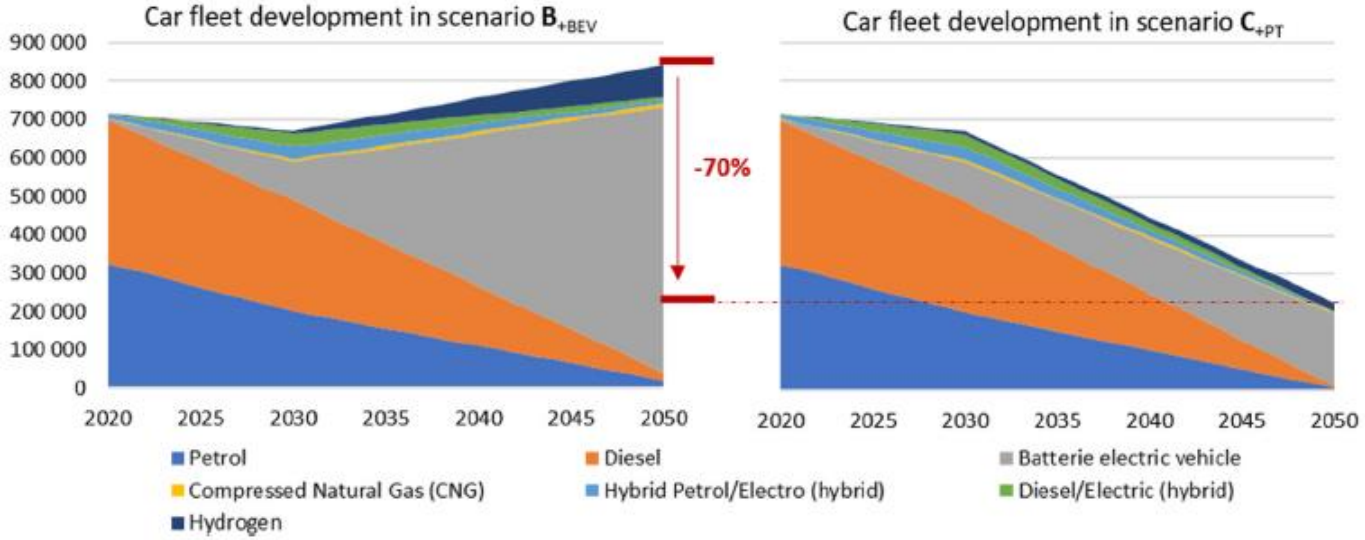
Projection for Private motorized vehicle



- Usage of conventional vehicles lowered in 2030 than 2020 due to change in NMT/PT usage
- Overall trips using private motorized vehicle will significantly be reduced in scenario C & D in 2050.

- As better PT & walking/cycle infrastructure and services make behavioral changes in society and if possible, people start walking/bicycle for short distances.

Car fleet type: Scenario B (EV) & C (PT) based on fuel type

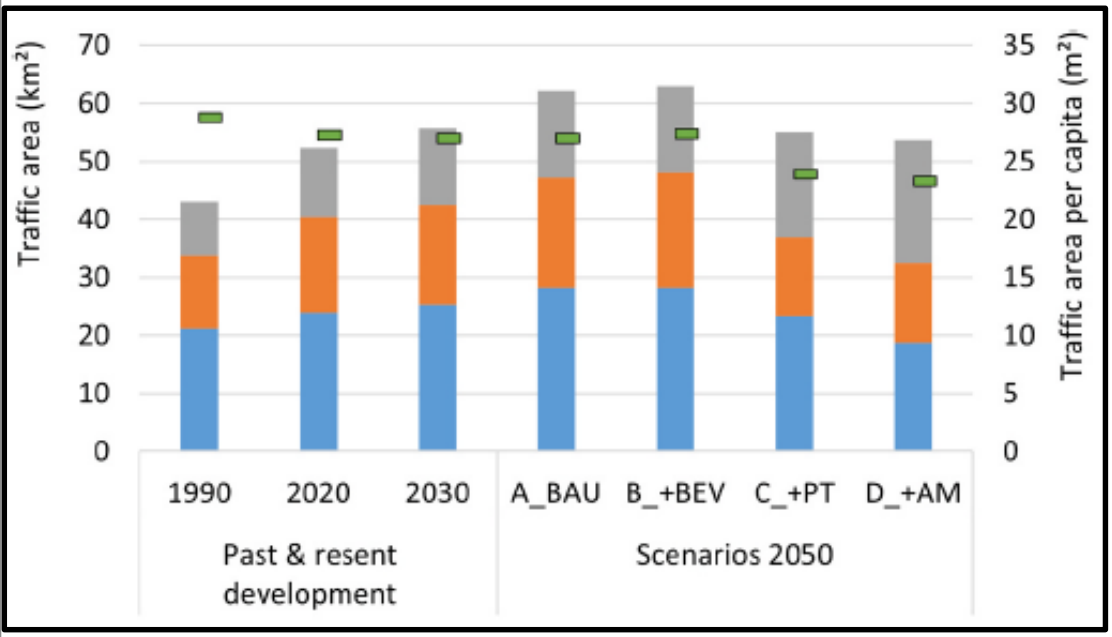


- Petrol & diesel fleet will be obsolete by 2050.
- 70% less vehicles expected in scenario compared to B.

- Hybrid cars will also be very less in no. in both the cases.

- In scenario B, hydrogen vehicles will rapidly increase after 2030

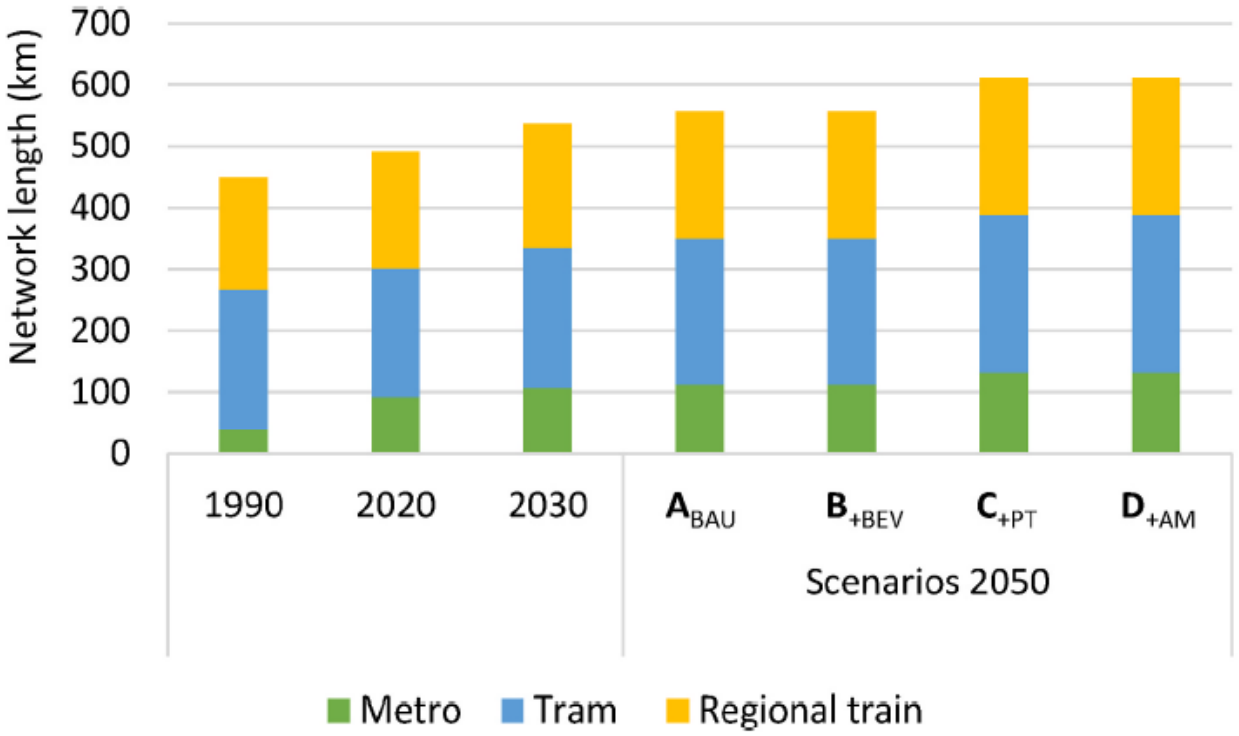
Infrastructure: Traffic area projection



Traffic area is sum of road, parking and walking/pedestrian services combined.

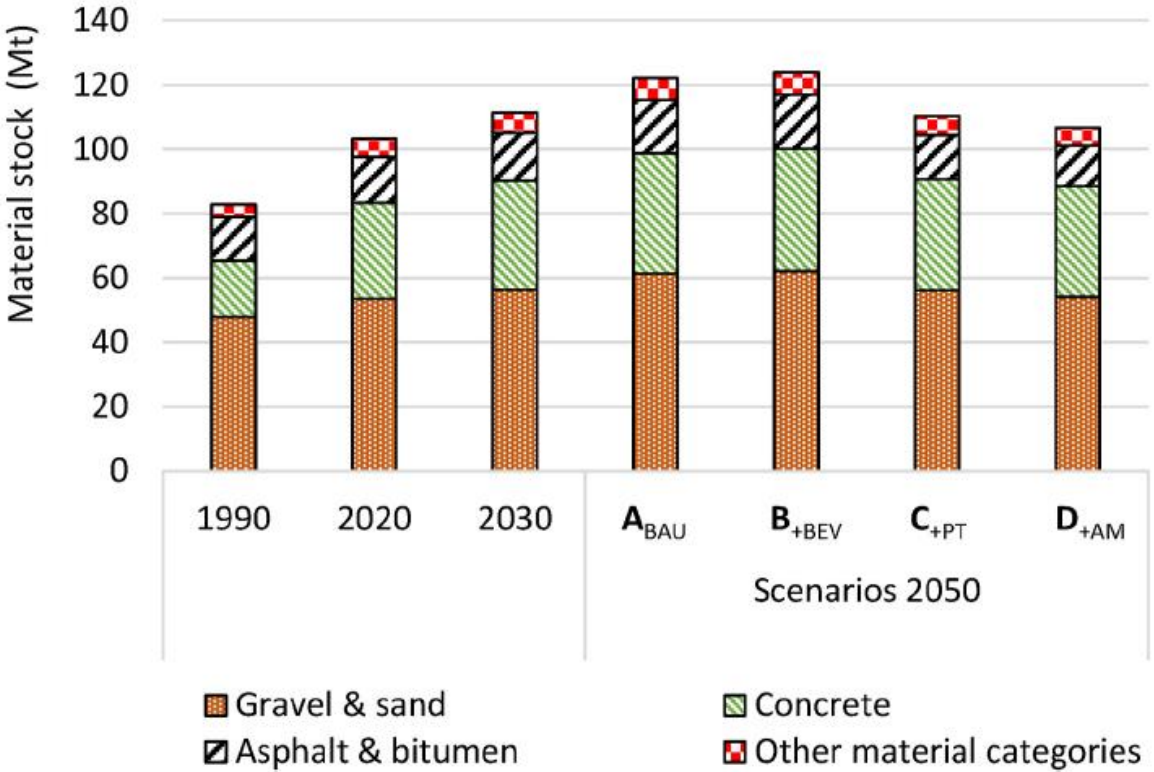
- Road & parking area will be reduced in scenario C & D compared to A & B
- Traffic area will also be reduced if we adopt PT or Active Mobility (AM) concept.

Rail Infrastructure



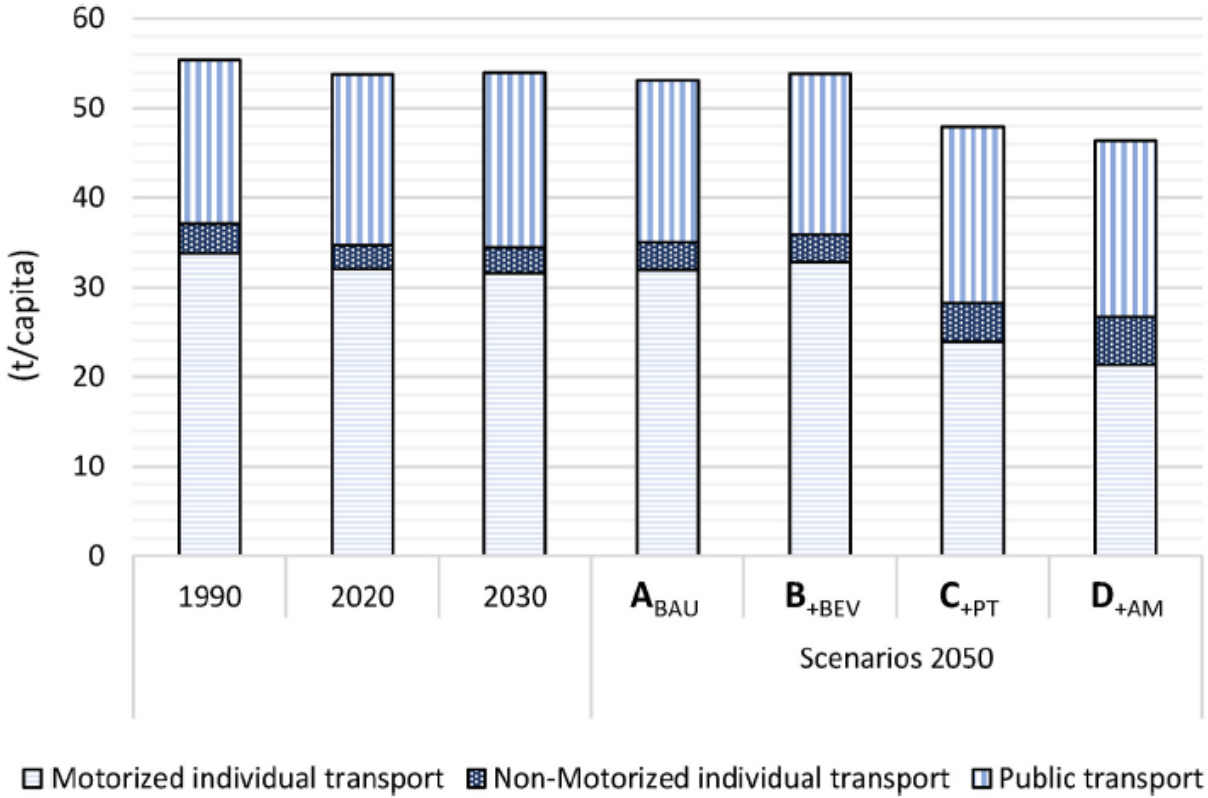
- Rail network length will be highest for scenario C & D.
- Same value for C & D affirms that public transport and NMT are supplementary to each other.

Transport construction material



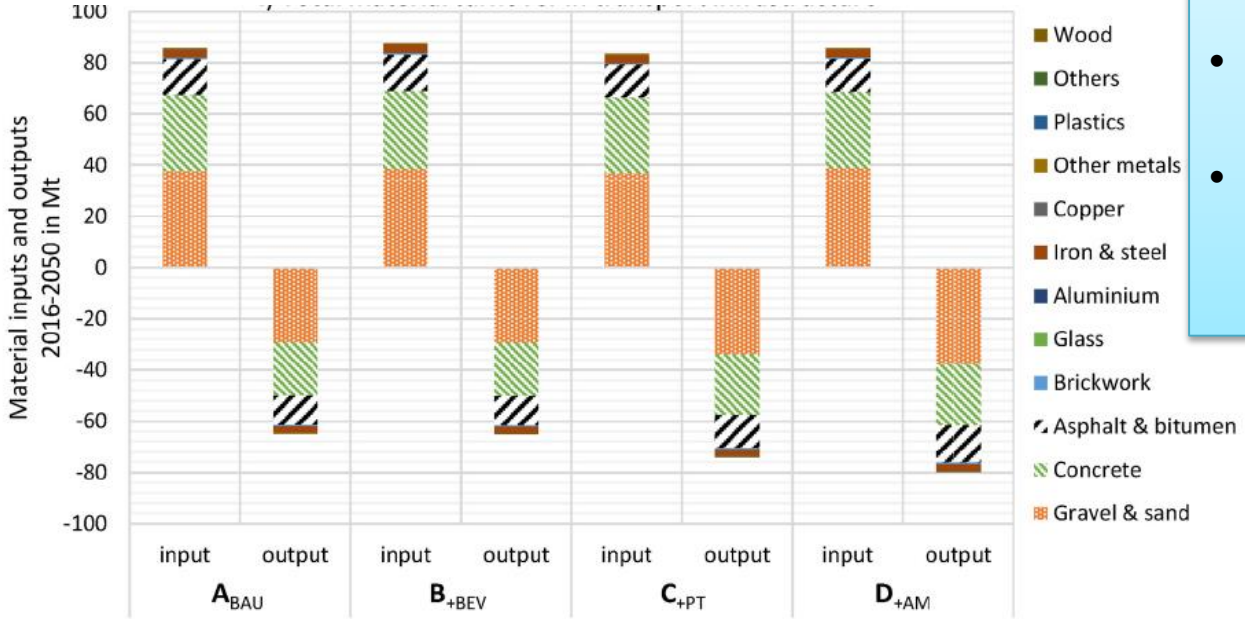
- With time, travel demand will increase so the material inflow.
- Scenario D will have least material usage.
- Scenario B will have higher value due to equal traffic area but increased material for vehicle manufacturing

Category wise material inflow per capita



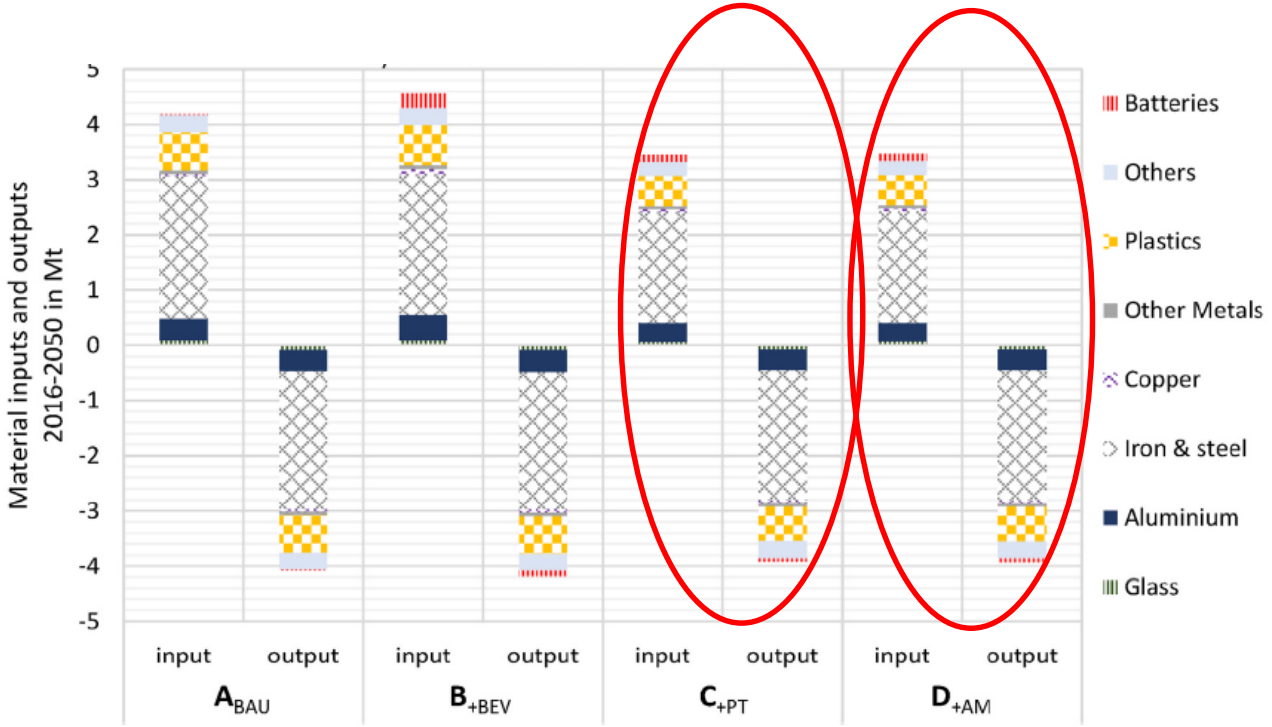
- Normalized values are taken for comparison.
- Significant reduction for scenario C & D in material for motorized individual transport(MIT) compared to A & B

Cumulative material in- and outputs into/from transport infrastructure



- Material flow for A & B scenario are similar also for C & D
- Output from C & D scenarios are higher than A & B
- Sand/gravel, concrete, asphalt and steel are major share of material

Cumulative material in- and outputs into/from vehicles



- Steel, aluminum & plastics are major component of input & output flow.
- Output for C & D scenario is higher than input due to dismantling of existing road infrastructure.

Conclusions of the study 1/3

- Choice of transport modes & the development of the motorization rate will play an important role in terms of the development of the material stock and annual resource consumption.

- A reduction in the consumption of primary raw materials is achievable by shifting from private motorized vehicle-based transport towards public transport and active mobility.

Conclusions of the study 2/3

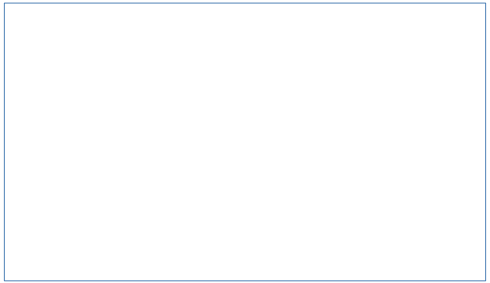
- If the existing system and modal split is retained, i.e. BAU scenario, resource consumption and the waste generated thereby will continue to rise although greenhouse emissions might well be mitigated due to fewer fossil-fuel based vehicles being used.

- The greatest savings potential results from a reduction in the private vehicle fleet and road infrastructure as well as from the associated decline in maintenance efforts.

Conclusions of the study 3/3

- To foster the transformation of the transport system towards less carbon and resource intensity and to avoid negative feedback loops, obsolete (road) infrastructure should be dismantled and converted for other purposes.

- Resource aspect should be considered for transport projects.



Final Conclusions

- Material flow analysis can also be used for evaluating various mode share scenarios.
- Apart from direct impact evaluation, material flow analysis should also be done for better understanding of transport projects in urban areas.
- Promoting public transport and NMT are also useful for resource conservation.

References

- Andreas Gassner, Jakob Lederer, Gerald Kovacic, Ursula Mollay, Christof Schremmer, Johann Fellner, Projection of material flows and stocks in the urban transport sector until 2050 – A scenario-based analysis for the city of Vienna, Journal of Cleaner Production, Volume 311, 2021, 127591, ISSN 0959-6526,

Thank You